

Unit1 Cloud Foundations

Cloud computing is the on-demand delivery of IT resources over the internet with pay-as-you-go pricing.

You only pay for what you use, and you can quickly access and scale resources like servers, storage, and databases as needed.

Cloud computing is built on the client-server model. Your device (the client) sends requests over the internet to remote servers, which process them and return results.

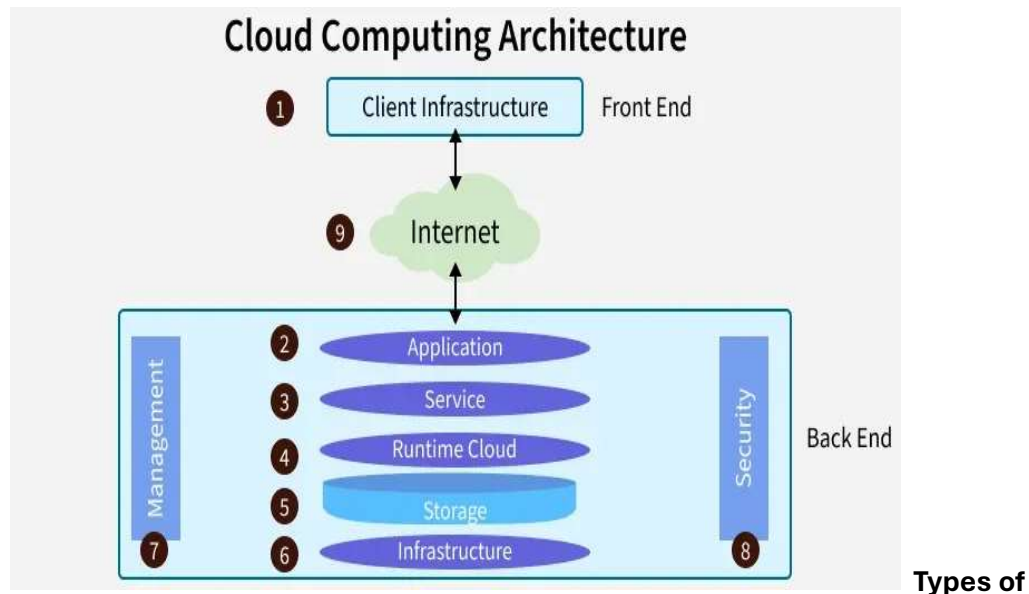
Problem That Cloud Computing Solves

Before the cloud, if a company wanted to launch a website or an application, it had to follow a slow and expensive process:

Cloud Computing Architecture

[Cloud computing architecture](#) refers to the components and sub-components required for cloud computing. These components typically refer to:

1. Front end (Fat client, Thin client)
2. Back-end platforms (Servers, Storage)
3. Cloud-based delivery and a network (Internet, Intranet, Intercloud)



Cloud Computing Services

4. The following are the types of Cloud Computing:
5. Types Of Cloud Computing Services
6. **1. Infrastructure as a Service (IaaS)**

7. Infrastructure as a Service ([IaaS](#)) is a type of cloud computing that gives people access to IT tools like virtual computers, storage, and networks through the internet. You don't need to buy or manage physical hardware. Instead, you pay only for what you use.

8. Benefits of using IaaS

9. **Flexibility and Control:** IaaS comes up with providing virtualized computing resources such as VMs, Storage, and networks facilitating users with control over the Operating system and applications.
10. **Reducing Expenses of Hardware:** IaaS provides business cost savings with the elimination of physical infrastructure investments making it cost-effective.
11. **Scalability of Resources:** The cloud provides in scaling of hardware resources up or down as per demand facilitating optimal performance with cost efficiency.

12. 2. Platform as a Service (PaaS)

13. Platform as a Service ([PaaS](#)) is a cloud computing model where a third-party provider offers the software and hardware tools needed to develop, test, and run applications. This allows users to focus on building their applications without worrying about managing servers or infrastructure.
14. **For example,** AWS Elastic Beanstalk is a PaaS offered by Amazon Web Services that helps developers quickly deploy and manage applications while AWS takes care of the needed resources like servers, load balancing, and scaling.

15. Benefits of using PaaS

16. **Simplifying the Development:** Platform as a Service offers application development by keeping the underlying Infrastructure as an Abstraction. It helps the developers to completely focus on application logic (Code) and background operations are completely managed by the AWS platform.
17. **Enhancing Efficiency and Productivity:** PaaS lowers the Management of Infrastructure complexity, speeding up the Execution time and bringing the updates quickly to market by streamlining the development process.
18. **Automation of Scaling:** Management of resource scaling, guaranteeing the program's workload efficiency is ensured by PaaS.

19. 3. Software as a Service (SaaS)

20. Software as a Service ([SaaS](#)) is a way of using software over the internet instead of installing it on your computer. The software is hosted by a company, and you can use it just by logging in through a web browser. You don't need to worry about updates, maintenance, or storage the provider takes care of all that.

21. A common example is Google Docs. You can write and share documents online without downloading any software.

22. Benefits of using SaaS

23. **Collaboration And Accessibility:** Software as a Service (SaaS) helps users to easily access applications without having the requirement of local installations. It is fully managed by the AWS Software working as a service over the internet encouraging effortless cooperation and ease of access.

24. **Automation of Updates:** SaaS providers manage the handling of software maintenance with automatic latest updates ensuring users gain experience with the latest features and security patches.

25. **Cost Efficiency:** SaaS acts as a cost-effective solution by reducing the overhead of IT support by eliminating the need for individual software licenses.

26.

1. Front End (User Interaction Enhancement)

The User Interface of Cloud Computing consists of 2 sections of clients. The Thin clients are the ones that use web browsers facilitating portable and lightweight accessibilities and others are known as Fat Clients that use many functionalities for offering a strong user experience.

2. Back-end Platforms (Cloud Computing Engine)

The core of cloud computing is made at back-end platforms with several servers for storage and processing computing. Management of Applications logic is managed through servers and effective data handling is provided by storage. The combination of these platforms at the backend offers the processing power, and capacity to manage and store data behind the cloud.

3. Cloud-Based Delivery and Network

On-demand access to the computer and resources is provided over the Internet, Intranet, and Intercloud. The Internet comes with global accessibility, the [Intranet](#) helps in internal communications of the services within the organization and the [Intercloud](#) enables interoperability across various cloud services. This dynamic network connectivity ensures an essential component of cloud computing architecture on guaranteeing easy access and data transfer.

Cloud Deployment Models

1. **Private Cloud:** The cloud infrastructure is operated solely for a single organization. It can be managed internally or by a third party and can exist on-premise or off-premise. It offers the highest level of security and control.

2. **Public Cloud:** The cloud infrastructure is owned and operated by a third-party cloud service provider (like AWS or Google) and is made available to the general public over the internet. It offers massive scalability and a pay-as-you-go model.
3. **Hybrid Cloud:** This model combines a private cloud with one or more public clouds, allowing data and applications to be shared between them. This offers flexibility, allowing companies to keep sensitive data on a private cloud while leveraging the scalable resources of a public cloud for other applications.
4. **Multi-Cloud:** A strategy where an organization uses a combination of clouds from two or more different public cloud providers. This helps avoid vendor lock-in and allows a company to use the "best-of-breed" service from each provider.

Introduction to Amazon Web Services

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform. Since its launch in 2006, AWS has revolutionized how individuals, companies and governments access technology services.

- Access computing power, storage, and databases on-demand
- Eliminate the need to buy, own, and maintain physical data centers
- Pay only for what you use (pay-as-you-go model)
- Flexible scaling based on business needs

AWS Global Infrastructure

Understanding AWS global infrastructure is essential for architecting resilient, low-latency applications.

Regions

- Physical locations worldwide where AWS clusters data centers
- Examples: us-east-1 (N. Virginia), ap-south-1 (Mumbai)
- Optimize latency by choosing regions close to your users
- Minimize costs (pricing varies by region)
- Meet data sovereignty and compliance requirements

Availability Zones (AZs)

- Multiple isolated locations within each region
- Each AZ contains one or more discrete data centers with redundant power, networking, and connectivity
- Ensures high availability and disaster recovery
- Prevents single points of failure

Edge Locations

- Smaller data centers in major cities worldwide
- Used by Amazon CloudFront (CDN) to cache content closer to end-users
- Reduces latency for faster content delivery

Shared Responsibility Model

Security is AWS's top priority, with responsibilities divided between AWS and customers.

AWS Responsibility (Security OF the Cloud)

- Protects cloud infrastructure, hardware, software, networking, and facilities
- Ensures data center security and compliance

Your Responsibility (Security IN the Cloud)

- Encrypt your data
- Configure network security (firewalls)
- Manage identity and access (IAM)
- Patch operating systems
- Implement application-level security